

Sample Script 3

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• import torch
• from torchvision.utils import save_image
• from diffusion import UNet, GaussianDiffusion, clip_normalize
•
• # Set device to GPU if available
• device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
•
• # Load the pre-trained U-Net model
• model = UNet().to(device)
• model.load_state_dict(torch.load("stable_diffusion_model.pth"))
•
• # Set the parameters for the diffusion process
• timesteps = 1000
• noise_scale = 0.05
•
• # Set the caption for the image
• caption = "Rick and Morty in a vat of acid"
•
• # Convert the caption to a tensor using CLIP
• with torch.no_grad():
•     text_embedding = clip_model.encode_text([caption]).to(device)
•
• # Initialize the image as noise
• image = torch.randn(1, 3, 512, 512).to(device)
•
• # Run the diffusion process
• diffusion = GaussianDiffusion(timesteps=timesteps, noise_schedule=noise_scale)
• for i, (noise, _) in enumerate(diffusion):
•     # Apply the diffusion step to the image
•     noise = noise.to(device)
•     image = model(image + noise, text_embedding)
•
• # Save the image every 100 steps
• if (i + 1) % 100 == 0:
•     image_normalized = clip_normalize(image)

```